Business Statistics

Variables and Scales of Measurement Flash Cards

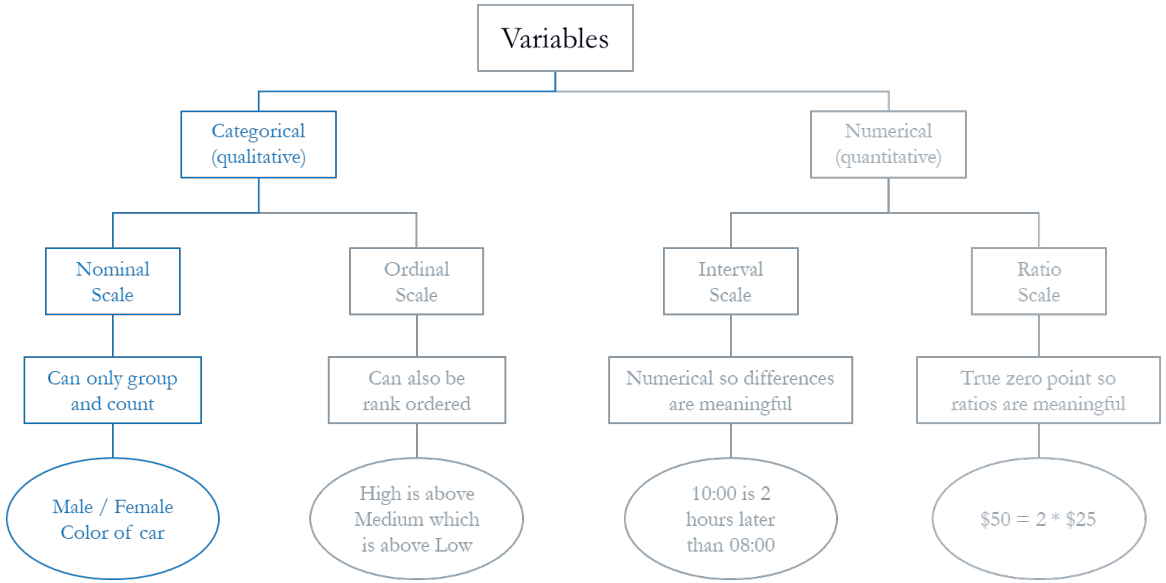
Some images and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Scales of Measurement
Variable: Eye color

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Eye color

- Categorical: Nominal
  - Categories with no ordering
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

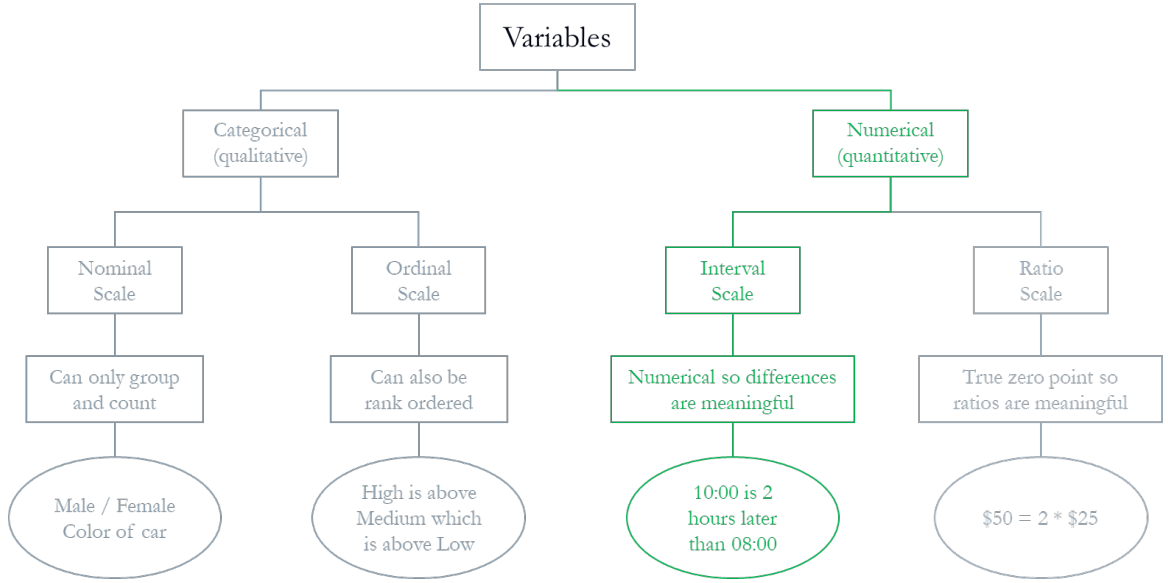


Scales of Measurement
Variable: Temperature in Fahrenheit

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Temperature in Fahrenheit

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
  - Numbers with meaningful differences
  - No true-zero point so 100 degrees is not 20% hotter than 80 degrees
- Numerical: Ratio



Scales of Measurement
Variable: Customer satisfaction rating

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

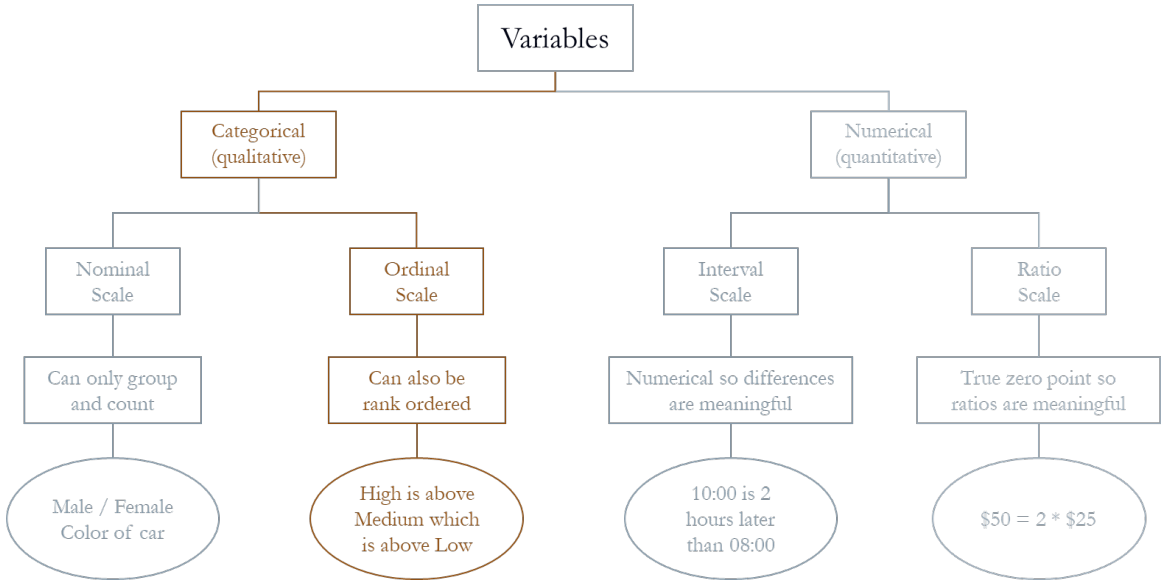
Scales of Measurement
Variable: Customer satisfaction rating

- Categorical: Nominal

- Categorical: Ordinal
  - Categories have inherent order
  - Unknown differences between levels

- Numerical: Interval

- Numerical: Ratio

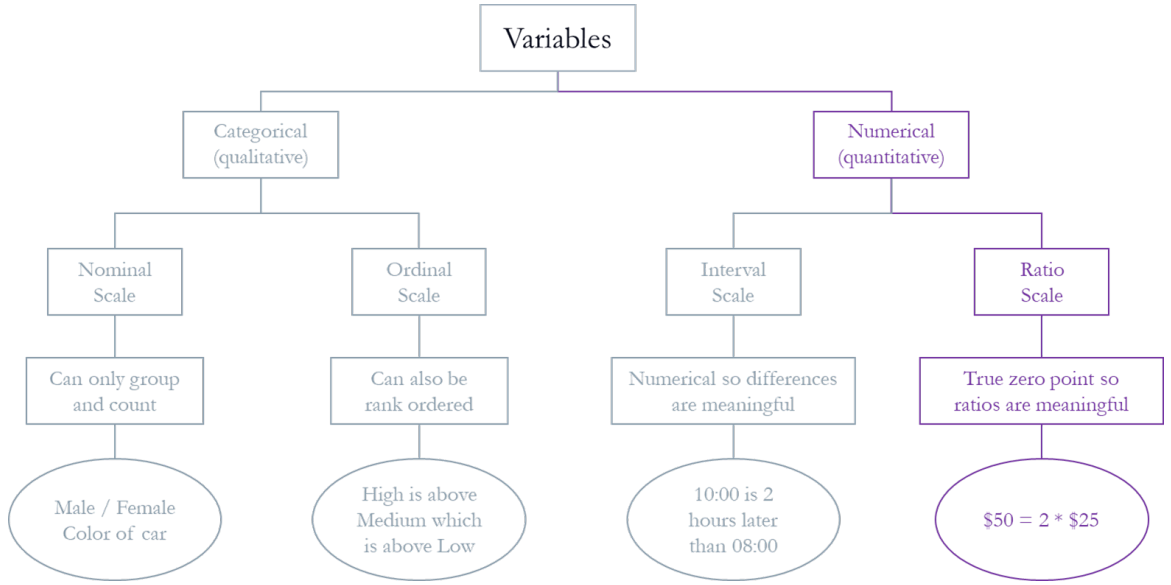


Scales of Measurement
Variable: Number of items purchased

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Number of items purchased

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio
  - Counts are always ratio data
  - 0 items means no items
  - 20 items is twice as many items as 10 items

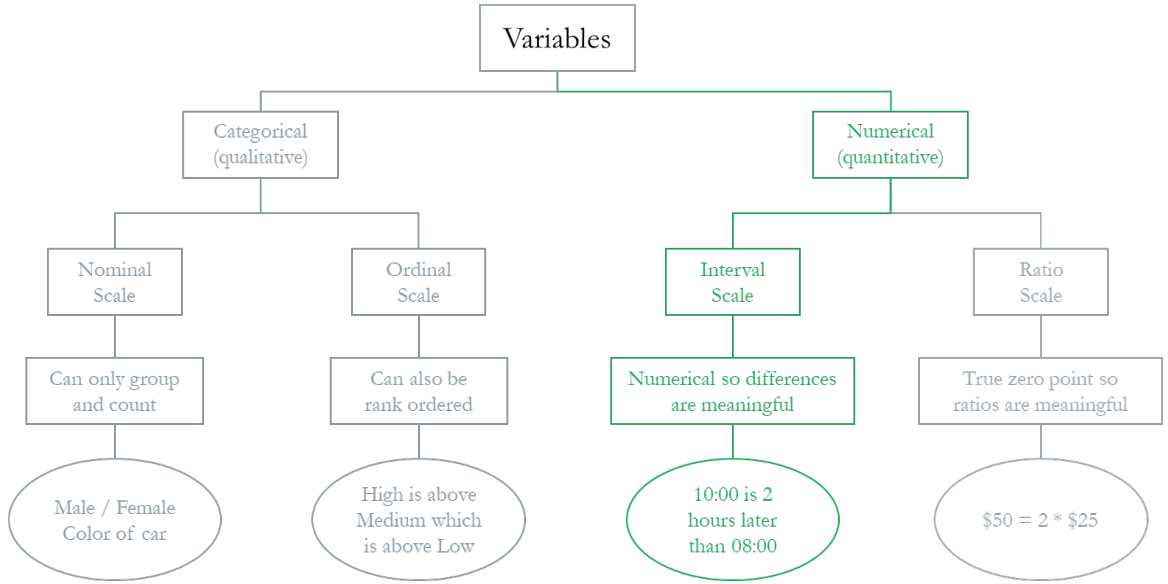


Scales of Measurement
Variable: Calendar year (e.g., 1990, 2000, 2020)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Calendar year (e.g., 1990, 2000, 2020)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
  - Equal spacing between years
  - 0000 does not mean “no years”
- Numerical: Ratio

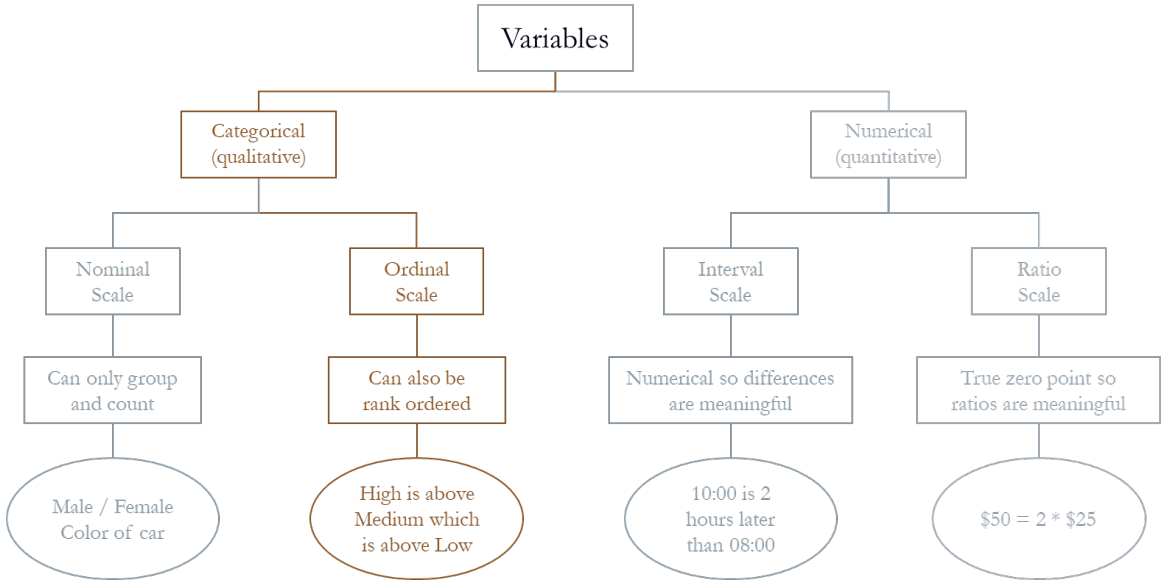


Scales of Measurement
Variable: Class (Freshman, Sophomore, Junior, Senior)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Class (Freshman, Sophomore, Junior, Senior)

- Categorical: Nominal
- Categorical: Ordinal
  - Categories have inherent order
- Numerical: Interval
- Numerical: Ratio

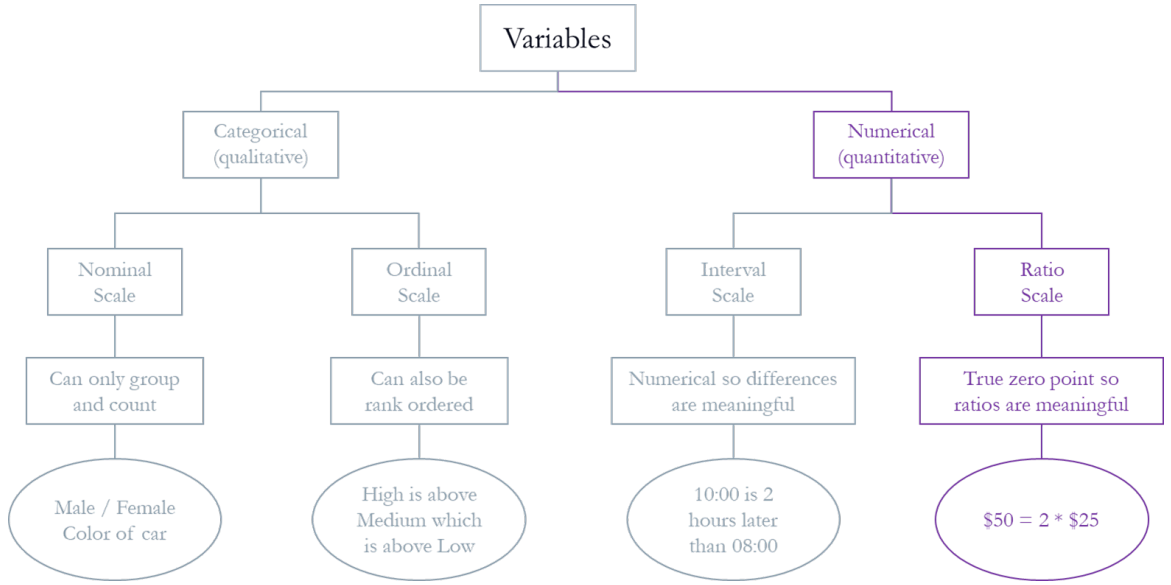


Scales of Measurement
Variable: Year in college (1st, 2nd, 3rd, 4th)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Year in college (1st, 2nd, 3rd, 4th)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio
  - Similar to counts
  - 0 means no years in college
  - 4th year is twice as long in college as 2nd year

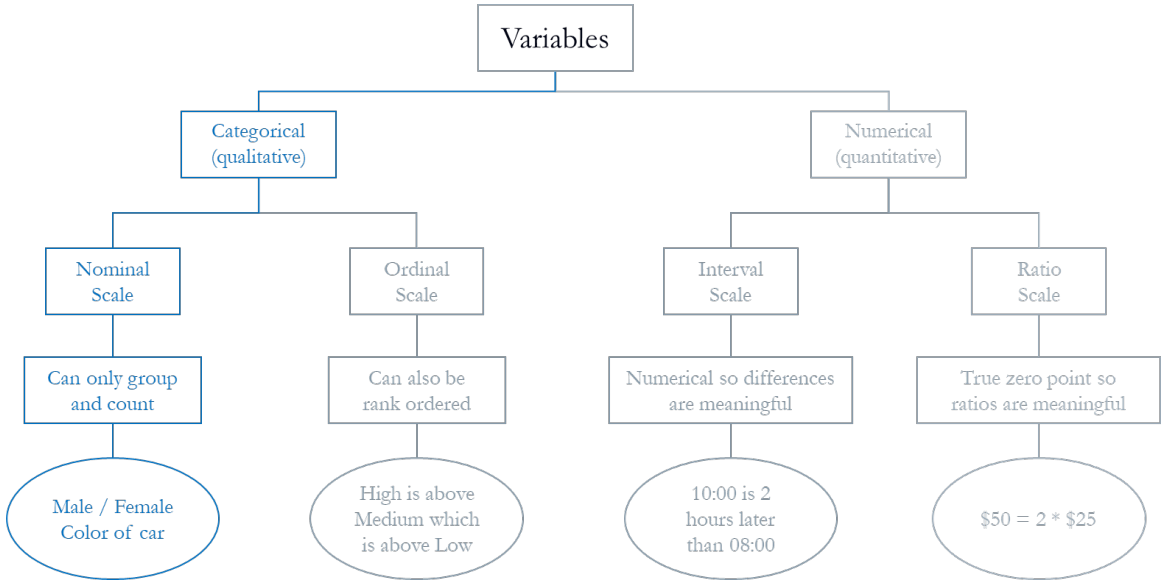


Scales of Measurement
Variable: Zip code

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Zip code

- Categorical: Nominal
  - Categories with no ordering
  - Numbers are labels only, arithmetic is meaningless
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

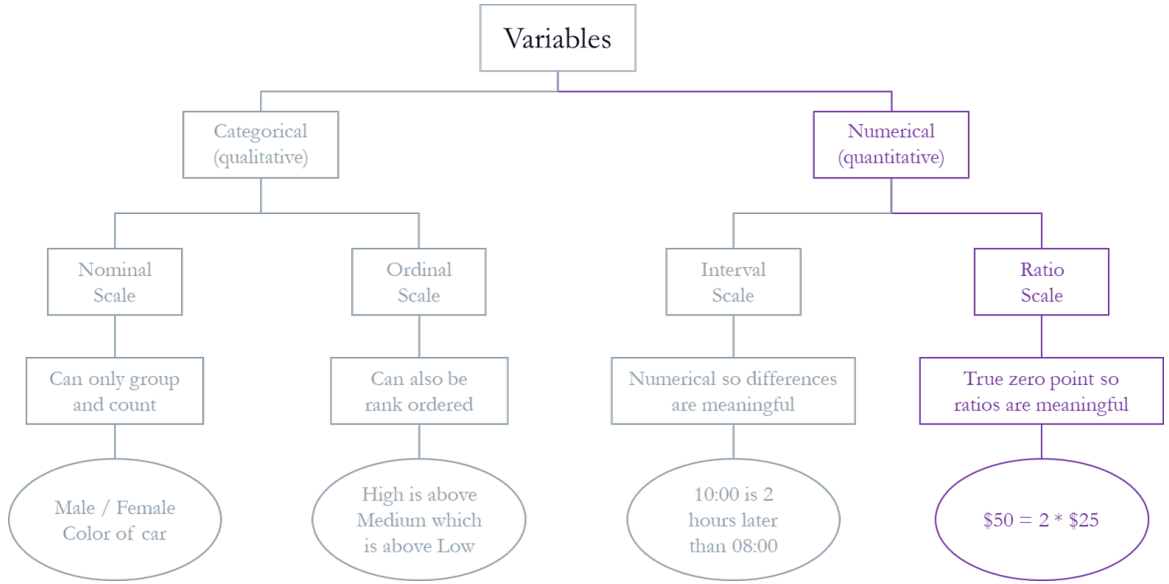


Scales of Measurement
Variable: Weight in pounds

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Weight in pounds

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio
  - Equal spacing between pounds
  - 0 pounds means no weight
  - 10 pounds is twice as heavy as 5 pounds

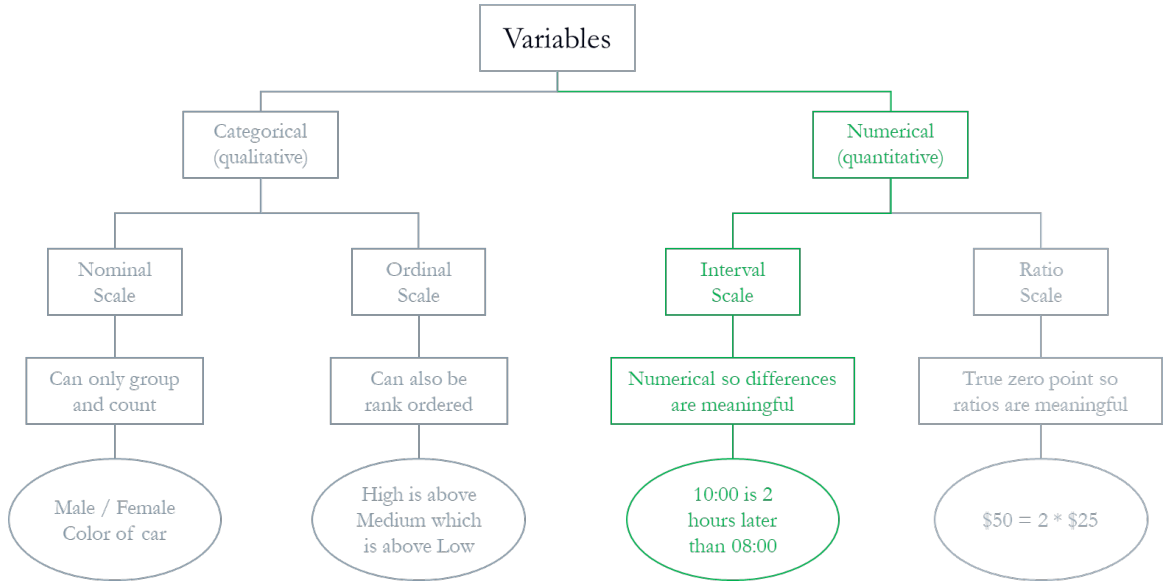


Scales of Measurement
Variable: Time of day (e.g., 3:00PM)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Time of day (e.g., 3:00PM)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
  - Equal spacing between hours
  - 00:00 does not mean “no time”
- Numerical: Ratio

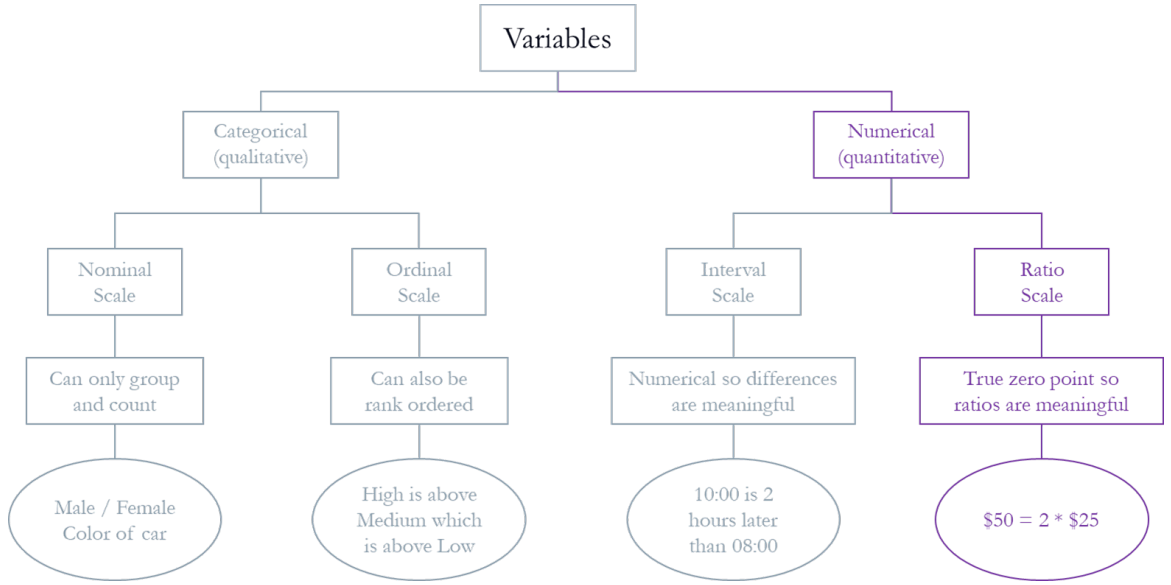


Scales of Measurement
Variable: Distance traveled (miles)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Distance traveled (miles)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio
  - Equal spacing between miles
  - 0 miles means no distance
  - 20 miles is twice as far as 10 miles

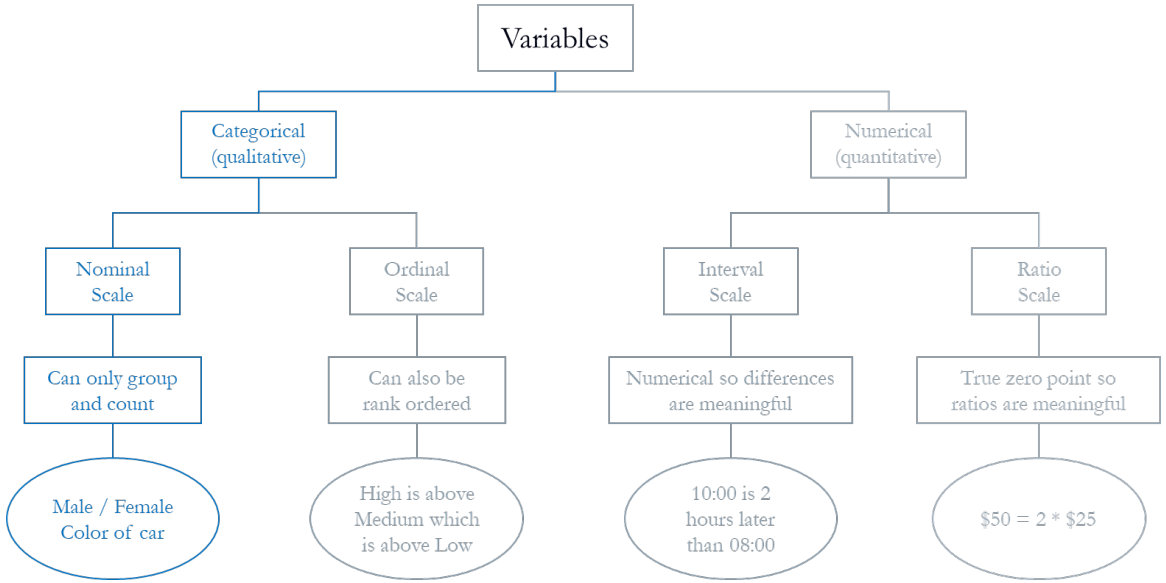


Scales of Measurement
Variable: Student ID number

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Student ID number

- Categorical: Nominal
  - Categories with no ordering
  - Numbers are labels only, arithmetic is meaningless
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio



Scales of Measurement
Variable: Salary coded into brackets (Low, Med, High)

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

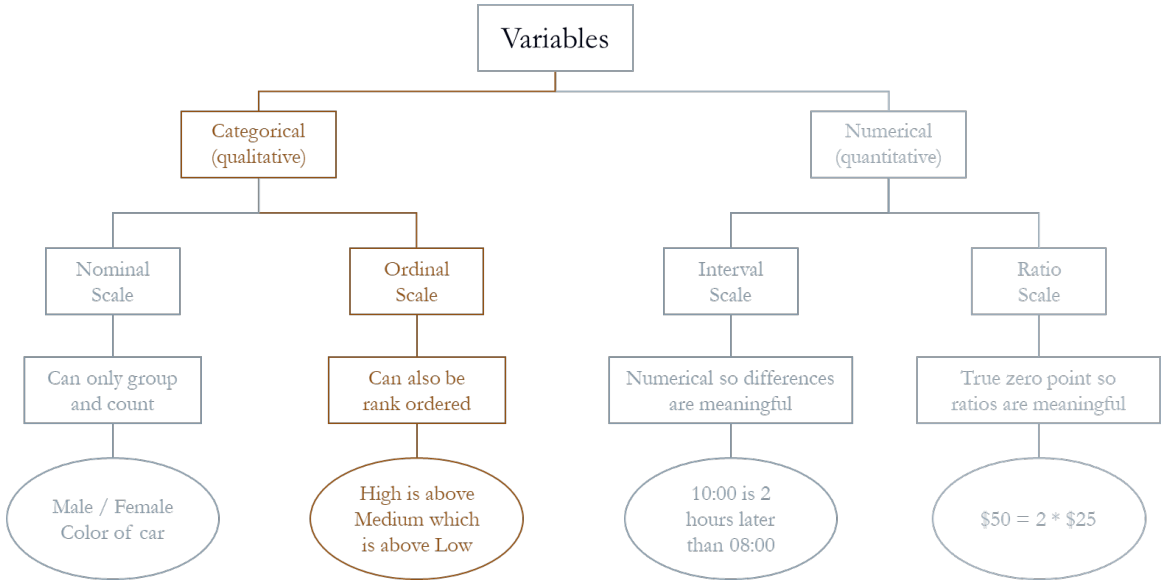
Scales of Measurement
Variable: Salary coded into brackets (Low, Med, High)

- Categorical: Nominal

- Categorical: Ordinal
  - Categories are ordered
  - Differences between levels unknown

- Numerical: Interval

- Numerical: Ratio



Scales of Measurement
Variable: Age in years

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio

Scales of Measurement
Variable: Age in years

- Categorical: Nominal
- Categorical: Ordinal
- Numerical: Interval
- Numerical: Ratio
  - Equal spacing between years
  - 0 years means no age
  - A 20-year-old is twice as old as a 10-year-old

